Does Pruning Transfer Across Datasets? A Comparative Study of Structured and Unstructured CNN Pruning

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Abstract—One supervised pipeline tests magnitude pruning of ResNet18 and VGG-16 on FloodNet, unmanned aerial vehicle (UAV), and Sen1Floods11, synthetic aperture radar (SAR). SAR labels use a water-fraction rule, and every number uses a split of labeled images. Global unstructured and filter-wise structured L_1 methods share a fine-tune. Three seeds (96 runs) give final-epoch accuracy, balanced accuracy, macro-F1, and AUC. On these two datasets, 90% unstructured pruning costs ResNet18 at most 0.03 macro-F1. At 50% structured pruning, FloodNet loses < 0.001 and Sen1Floods11 loses 0.067 macro-F1; the gap grows with the ratio. A lower fine-tuning learning rate does not close it. One dense VGG-16 SAR seed fails (AUC near 0.858), and structured runs collapse. Recheck a sparsity budget on the other dataset.

Index Terms—network pruning, model compression, flood detection, synthetic aperture radar (SAR), optical remote sensing, FloodNet, Sen1Floods11

I. Introduction

Flood maps are needed within hours, often on a small UAV, an edge box, or a nanosatellite that cannot host a large network [1], [5]. Magnitude pruning removes small weights or whole filters and then fine-tunes the network [6], [7], [13]. On optical benchmarks this often keeps accuracy at high sparsity [12]. Those tests are usually run on one dataset. A model compressed for optical UAV images may later be asked to read Sentinel-1 SAR, which is the sensor that still sees the ground through cloud [2].

This study asks one question:

> Does a heavily pruned classic CNN hold its accuracy on both an optical UAV flood dataset and a SAR flood dataset, or does one dataset break down while the other does not?

We do not claim a general law about sensors. FloodNet and Sen1Floods11 also differ in size, resolution, class balance, and how the labels were made. The controlled part of the study is the pipeline: the same architectures, input size, augmentation, sampler, optimizer, schedule, pruning operators, ratios, and fine-tuning budget, with only the dataset changed.

The backbones are ResNet18 (11.18M parameters) [17] and VGG-16 (134.3M) [18]. Sen1Floods11 is labeled per pixel. We turn each hand-labeled mask into one image label with the rule in Section III. That positive class means "at least 1% of the valid pixels are water." The mask includes permanent water (rivers and lakes). We did not remove it, so the SAR label is not a confirmed flood label. Contributions.

- A 96-run study (two architectures, two datasets, two pruning families, several ratios, three seeds) with the training pipeline held fixed.
- An explicit rule that turns Sen1Floods11 water masks into image-level labels so they can be scored like FloodNet. The code and split lists are released (see the conclusion).
- On these two datasets, unstructured pruning of ResNet18 costs at most 0.03 macro-F1 at 90% sparsity, while 50% structured pruning costs less than 0.001 on FloodNet and 0.067 on Sen1Floods11. A lower fine-tuning learning rate, within the range we tested, does not remove that gap.
- VGG-16 as an instability case: one dense SAR seed fails at the final-epoch threshold while AUC stays near 0.858, and structured runs collapse on both datasets. This is consistent with a serial network that has no skip connections and no batch normalization (BatchNorm); we do not claim that missing skips are the only cause.

II. Related Work

A. Datasets and Flood Classifiers

FloodNet [1] is UAV imagery from Hurricane Harvey. Its Track-1 classification split has 398 labeled training images (51 flooded, 347 non-flooded). The official validation and test classification labels were withheld, so any labeled

score has to come from those 398 images. Sen1Floods11 [2] has 512×512 Sentinel-1 chips from 11 flood events, with 446 hand-labeled chips. The mask values are water, non-water, and invalid. It is a segmentation dataset. CNN flood mapping on Sentinel-1 is usually done at pixel level [3]; we instead score a single label per chip so the pruning comparison matches FloodNet.

Khose et al. [4] report 96.70% test accuracy for a semi-supervised ResNet18 (11.6M parameters) on FloodNet, using pseudo-labels [20]. We use supervised training only, so pruning is not mixed with pseudo-label noise. Khan et al. [5] report 96.43% accuracy and 86.36% F1 score for Flood-ResNet50 (98 megabytes (MB), 4.3 billion floating-point operations (FLOPs)) on a Jetson Nano. Their F1 is not our macro-F1. Both published scores use the hidden FloodNet test set. Our dense ResNet18 reaches 97.0% accuracy and 0.929 macro-F1 on a 199-image split of the labeled training images. Those numbers are not directly comparable.

B. Pruning

Unstructured magnitude pruning zeros individual weights [6]. It needs sparse kernels to become faster. Structured pruning removes whole filters [7], which runs on ordinary hardware, but it is a coarser choice. Channel pruning [8] and Taylor-based filter pruning [9] are other structured criteria; we use the L_1 norm because it is simple and identical on both datasets. Later work argues that pruning mostly reveals a trainable subnetwork [10] or a better architecture [11], and that papers test these ideas on a few optical benchmarks with incompatible protocols [12], [13]. Compression can also hurt some examples much more than the average accuracy shows [14]. That matters here: flooded images are 12.8% of FloodNet and 66.4% of our Sen1Floods11 labels, so accuracy alone can hide a class that collapsed.

Pruning has been used inside remote sensing, but on one dataset at a time: channel pruning for optical scene classification [15], and filter pruning plus distillation for SAR target recognition [16]. We did not find a study that freezes the flood-classification pipeline and changes only the dataset. That is the comparison we run.

III. Experimental Setup

Fig. 1 is the whole study. Table I lists the data actually used.

A. Labels and Inputs

FloodNet images are 4000×3000 (width×height). We resize them to 400×300, which keeps the 4:3 frame. The 398 labeled images are split 50/50, stratified, with data seed 42 (199/199). This follows the public baseline's use of the labeled training images [4], because the official test labels are hidden. The validation half contains about 25 flooded images.

TABLE I
Data used in this study. Sizes are width×height. RGB is red, green, and blue. VV is vertical-transmit vertical-receive and VH is vertical-transmit horizontal-receive.

	FloodNet	Sen1Floods11
Sensor	UAV, RGB	Sentinel-1, VV+VH
Original size	4000×3000	512×512
Network input	400×300	400×300
Labeled images	398	441
Train / validation	199 / 199	220 / 221
Positive class	12.8% flooded	66.4% water
How labeled	image label	mask, τ=1%

For each Sen1Floods11 chip we compute

$$y = \mathbf{1}\left[\frac{|\{p : m_p = 1\}|}{|\{p : m_p \neq -1\}|} \geq \tau\right], \qquad \tau = 0.01, \tag{1}$$

and drop chips with no valid pixels. A positive chip has at least 1% water among valid pixels. At this threshold a river is enough to count as positive, and permanent water was not removed, so y=1 means "contains water," not "flooded." The same rule yields 441 chips (293 positive / 148 negative), split 50/50 stratified with the same data seed (220/221). The split is by chip, not by flood event, so neighboring chips from one event can fall on both sides. We treat that as a limitation, not as a solved robustness check.

SAR backscatter is clipped to $[-50, 1]$ decibels (dB), scaled to $[0, 1]$, and stacked as $(\text{VV}, \text{VH}, (\text{VV}+\text{VH})/2)$. The same clip, scale, and ImageNet normalization are used for training and validation. Chips are 512×512 and are resized to 400×300 so both datasets share one input tensor. That changes the SAR aspect ratio from 1:1 to 4:3.

The class imbalance runs in opposite directions (Table I). We therefore report balanced accuracy and macro-F1 with accuracy and AUC, and we compare each pruned model only with its own dense baseline.

B. Training

Every dense run and every fine-tune uses ImageNet initialization [19], a 2-logit head, unweighted cross-entropy, and a class-balanced weighted sampler. The only augmentation is random horizontal and vertical flips. The optimizer is stochastic gradient descent (SGD) with learning rate 0.01, momentum 0, and weight decay 0. This recipe is unusual. We keep it because the study holds one recipe fixed; we do not claim it is the best optimizer for SAR. Each run lasts 50 epochs. We score the last epoch, with no early stopping and no best-validation checkpoint. Seeds 42, 124, and 2024 change initialization, sampling, and augmentation. The split does not change. Batch size is 64 for ResNet18 and 16 for VGG-16, and it is the same on both datasets. ResNet18 BatchNorm layers stay in train mode during fine-tuning and in eval mode at test time. Torchvision VGG-16 has no BatchNorm. All 96 runs were executed on Kaggle. That session attaches

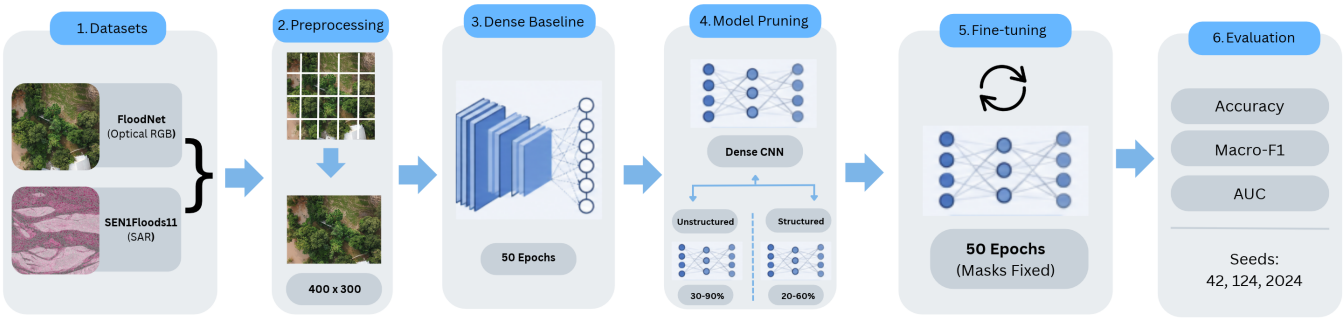


Fig. 1. Study pipeline. Both datasets follow the same steps. Every pruned run starts from the dense baseline of the same seed. Architecture, resolution, augmentation, sampler, optimizer, schedule, operators, ratios, and fine-tuning budget stay fixed. Only the dataset changes.

FloodNet (optical UAV)

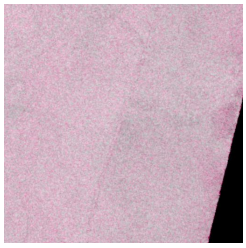


(a) Flooded



(b) Non-flooded

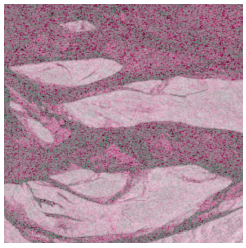
Sen1Floods11 (Sentinel-1 SAR)



(c) Bolivia_360519



(d) Mask, 1.1% water



(e) India_1017769



(f) Mask, 56.7% water

Fig. 2. Examples from the two datasets. (a) and (b) are FloodNet optical UAV images. (c) and (e) are Sen1Floods11 chips as the three-channel composite the network sees (VV, VH, and their mean). (d) and (f) are the water masks of those same chips: white is water, black is non-water, gray is invalid. Water among valid pixels is 1.1% on Bolivia_360519 and 56.7% on India_1017769. Eq. (1) turns each mask into one image label.

two NVIDIA Tesla T4 graphics processing units (GPUs), the free accelerator named GPU T4 x2, and each model was trained on one of them. Throughout, $\pm$ is the sample standard deviation over the three seeds. The logs store the population standard deviation; we rescale by $\sqrt{3/2}$.

C. Pruning

Each pruned run loads the dense checkpoint of the same seed.

- Global unstructured L_1 [6]. Weights of every convolution and linear layer are ranked together, and the smallest fraction $r \in \{0.5, 0.9\}$ is zeroed. Biases are not pruned.
- Layer-wise structured L_1 [7]. In every convolution with more than 8 output channels, and every linear layer with more than 8 outputs, the fraction $r \in \{0.3, 0.5, 0.7, 0.9\}$ of filters (or rows) with the smallest L_1 norm is masked. The 2-logit classifier is never pruned.

The two families are not the same intervention. Structured pruning uses one ratio in every eligible layer. Unstructured pruning spends the budget wherever the global magnitude is smallest. They differ in granularity and in how sparsity is spread across layers.

Masks stay fixed for 50 fine-tuning epochs at learning rate 0.01. A control repeats ResNet18 at 50% and 90%, and VGG-16 at 70%, on both datasets, at learning rate 0.001 for two seeds (42 and 124) and the same 50 epochs. That control is a lower rate only. It is a short schedule with no momentum, so it cannot show that the gap survives

TABLE II
ResNet18, final epoch, mean ± sample standard deviation (std) over 3 seeds. Par. is remaining stored parameters in millions, not physical channel-removed size. mF1 is macro-F1. Unstruct. is unstructured. Fine-tuning learning rate 0.01. The best mean in each metric column is in bold.

			FloodNet		Sen1Floods11	
Method	Ratio	Par.	mF1	AUC	mF1	AUC
dense	0%	11.18	.929±.013	.991±.003	.827±.011	.908±.016
unstruct.	50%	5.59	.917±.012	.995±.002	.833±.008	.912±.014
unstruct.	90%	1.13	.911±.011	.989±.002	.799±.016	.888±.021
structured	30%	7.82	.925±.014	.986±.015	.768±.037	.858±.019
structured	50%	5.59	.928±.020	.989±.005	.759±.008	.841±.012
structured	70%	3.37	.857±.077	.930±.063	.734±.034	.834±.015
structured	90%	1.13	.798±.078	.930±.022	.577±.047	.763±.032

TABLE III
ResNet18 accuracy (Acc) and balanced accuracy (BAcc), same runs as Table II. Means over 3 seeds, sample std. The best mean in each metric column is in bold. Ties are both bold.

		FloodNet		Sen1Floods11	
Method	Ratio	Acc	BAcc	Acc	BAcc
dense	0%	.970±.005	.914±.020	.840±.011	.840±.009
unstruct.	90%	.960±.005	.920±.012	.815±.016	.811±.016
struct.	50%	.970±.009	.909±.013	.780±.007	.767±.010
struct.	90%	.893±.072	.836±.058	.578±.047	.656±.044

a stronger recipe. We report its mean and no standard deviation, because two seeds do not support one. The full grid is 12 dense, 48 structured, 24 unstructured, and 12 learning-rate runs.

Two implementation facts change how the parameter counts should be read. First, structured pruning masks output filters only. The next layer still has its input channels, and BatchNorm scale and bias on a masked ResNet filter are not removed. A zeroed convolution followed by BatchNorm can still emit a constant, so the channel is not physically gone. The horizontal axis of Fig. 3 is the count of remaining stored parameters, including unpruned biases and BatchNorm terms. It is not a FLOPs count and not the size after deleting channels. At 50% both families report 5.59M remaining parameters for that reason; a network with the channels actually deleted would be smaller. Second, ResNet18's parameter-free identity shortcuts are not pruned. Its 1×1 projection shortcuts are convolutions with 128, 256, or 512 outputs, so the same structured rule does prune them.

IV. Results and Discussion

A. ResNet18

Tables II and III and Fig. 3 are the main result. Drops below are the difference of the unrounded means, then rounded to three decimals.

Unstructured pruning transfers on these two datasets. At 90% sparsity (1.13M remaining parameters) macro-F1 falls by 0.018 on FloodNet ($0.929 \to 0.911$) and by 0.028 on Sen1Floods11 ($0.827 \to 0.799$). AUC moves by at most 0.020. The dense SAR model is only moderately strong (accuracy 0.840, AUC 0.908), so this is stability around a modest baseline, not a claim that SAR classification is solved.

TABLE IV
ResNet18 structured pruning, macro-F1 (mF1). lr is the learning rate. The 0.01 column is 3 seeds. The 0.001 column is the mean of 2 seeds and has no std. The best mean in each column is in bold. The lower rate does not restore Sen1Floods11.

Dataset	Ratio	mF1, lr 0.01	mF1, lr 0.001
FloodNet	50%	.928±.020	.914
FloodNet	90%	.798±.078	.699
Sen1Floods11	50%	.759±.008	.718
Sen1Floods11	90%	.577±.047	.609

Structured pruning does not transfer. On FloodNet, 50% filter pruning changes macro-F1 by less than 0.001 ($0.9288 \to 0.9283$). The 70% and 90% FloodNet rows are already noisy (sample std about 0.08), so the supported statement is "no clear loss up to 50%," not "no loss at every ratio." On Sen1Floods11 the same operator lowers macro-F1 by 0.058 at 30%, 0.067 at 50%, 0.092 at 70%, and 0.249 at 90%. AUC goes from 0.908 to 0.763 at 90%. The 50% drop is several times the seed spread of that row (sample std 0.008). With only three seeds we do not report a formal significance test. We also do not say that Sen1Floods11 "needs" the removed filters. The equal-parameter plot is not a pure comparison of structured versus unstructured pruning, because the two methods allocate sparsity differently and neither count is a channel-deleted model.

Table IV only answers a narrow question: is the SAR drop an accident of using learning rate 0.01 rather than 0.001? At 50% the lower rate is worse on Sen1Floods11 (0.718 versus 0.759). At 90% it is slightly better (0.609 versus 0.577) and still 0.218 below the dense mean of 0.827. On FloodNet the lower rate is close at 50% and worse at 90%. The gap is not explained by this one downward change of learning rate. A higher rate, momentum, or a longer schedule was not tested.

B. VGG-16

Table V is an instability table, not a second clean curve. On Sen1Floods11 the dense accuracy is 0.661 ± 0.282. Seed 42 predicts the negative class on every validation chip (accuracy 0.335, balanced accuracy 0.500, macro-F1 0.251) while its AUC is 0.858. Seeds 124 and 2024 reach macro-F1 0.798 and 0.814, with AUC 0.861 and 0.855. The failed seed still ranks the chips. What fails is the final-epoch decision threshold. We have not re-fit that threshold on the training set, so this reading is from the AUC, not from a calibration experiment. The same pattern appears in the FloodNet learning-rate row: macro-F1 0.289 with AUC 0.922.

The dagger matches the table. It marks every row with at least one constant-predictor seed, including dense and

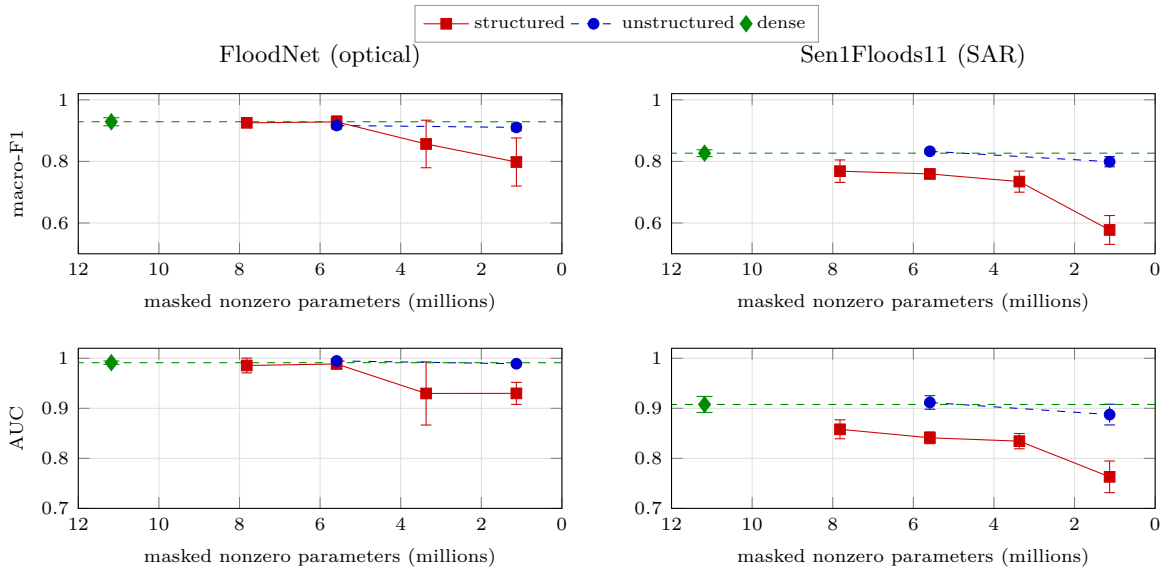


Fig. 3. ResNet18 macro-F1 (top) and AUC (bottom) against masked nonzero parameters. Squares, solid: structured. Circles, dashed: unstructured. Green diamond and dashed line: dense mean. Error bars: sample std over 3 seeds. This axis is not a physical compression axis (Section III).

TABLE V
VGG-16, final epoch, sample std over 3 seeds. The last row is the mean of 2 seeds. † means at least one seed has balanced accuracy 0.5 (a constant prediction). The best mean in each metric column is in bold. Ties are both bold.

		FloodNet		Sen1Floods11	
Method	Ratio	mF1	AUC	mF1	AUC
dense	0%	.924±.007	.990±.000	.621±.321†	.858±.004
unstruct.	50%	.924±.007	.990±.001	.601±.304†	.849±.006
unstruct.	90%	.943±.018	.980±.024	.597±.300†	.846±.017
struct.	30%	.386±.244†	.737±.043	.586±.290†	.834±.006
struct.	50%	.803±.093	.940±.033	.519±.233†	.788±.009
struct.	70%	.621±.078	.675±.229	.300±.086†	.502±.276
struct.	90%	.230±.205†	.500±.000	.300±.086†	.500±.000
lr 0.001	70%	.289†	.922	.325†	.700

unstructured SAR, not only the structured rows. All three seeds collapse (balanced accuracy 0.5 and AUC 0.5) for structured pruning at 90% on FloodNet and at 70% and 90% on Sen1Floods11. Other daggered rows still have AUC well above 0.5. Structured 30% on FloodNet (macro-F1 0.386) is worse than 50% (0.803) because a collapsed seed pulls the mean down. We do not call that curve monotone.

At 90% structured pruning the stem keeps 6 of 64 filters (then 13 of 128, 26 of 256, 51 of 512, and 410 of 4096 in the fully connected layers), because the operator removes round$(0.9\,n)$ filters in each layer. VGG-16 has no identity path around those layers, and this torchvision VGG-16 also has no BatchNorm and a large fully connected head. Those facts are consistent with the collapses. They are not a controlled proof that missing skip connections are the cause. ResNet18 degrades instead of collapsing at the same ratios; its identity shortcuts are unpruned, while its projection shortcuts are pruned.

On FloodNet, 90% unstructured VGG-16 reaches macro-F1 0.943 ± 0.018 against a dense 0.924 ± 0.007. The difference sits inside the seed noise. We do not claim that pruning improves VGG-16.

C. What the Comparison Supports

On these two datasets, a 90%-sparse unstructured ResNet18 stays within 0.03 macro-F1 of its dense model on both FloodNet and Sen1Floods11. Structured filter pruning does not: FloodNet loses less than 0.001 macro-F1 at 50%, and Sen1Floods11 loses 0.067, with a larger drop at higher ratios. That SAR drop is not removed by changing the fine-tuning learning rate from 0.01 to 0.001. A budget measured on one of these datasets should be measured again on the other, with balanced accuracy, macro-F1, and AUC together. This is not yet a sensor-only result. The datasets differ in more than the camera.

V. Limitations

The pipeline is fixed, so the gap can be tied to the dataset as a whole. It cannot be tied to SAR versus optical alone. FloodNet and Sen1Floods11 differ in count (398 vs. 441), resolution (4000×3000 UAV frames vs. 512×512 chips at 10 meters), imbalance (12.8% vs. 66.4% positive), geography, and labels (a human image label vs. Eq. (1)). The SAR split is random over chips, so chips from the same event can leak across the split. The positive SAR class still contains permanent water, and we did not sweep τ.

Three seeds and a validation set of 199 images (about 25 flooded) or 221 chips can show a 0.067 drop. They cannot support small differences. Both networks are pre-2016 CNNs with ImageNet weights, which favor the optical input. The same ratios may not hold for MobileNet, EfficientNet, or a transformer. VGG-16's SAR failures are also tied to a no-BatchNorm model and a final-epoch threshold. Structured channels are masked, not deleted, so we report no speedup. Unstructured sparsity does not get faster without a sparse kernel.

VI. Conclusion

We compared one pruning pipeline on FloodNet and on Sen1Floods11. Unstructured magnitude pruning of ResNet18 transfers across these two datasets, within 0.03 macro-F1 at 90% sparsity. Structured filter pruning does not: the 50% cost is under 0.001 macro-F1 on FloodNet and 0.067 on Sen1Floods11, and a lower fine-tuning learning rate does not repair it. VGG-16 is a warning, not a second confirmation. One dense SAR seed fails at the threshold while AUC stays near 0.858, and structured runs collapse on both datasets. Re-check structured compression on every target dataset. A paired optical/SAR test on the same chips, an event-wise SAR split, a sweep of τ, more seeds, and a count of parameters after the channels are actually removed are left to future work. Code and the fixed split lists: https://github.com/AbdulRauf20/flood-image-classification-deep-learning/tree/main/cross_dataset_pruning.

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